

Active InferAnt Stream 010.2 ~ plain_text too: Learning Paths, Application

ActiveInferAntStream | plain_text too: Learning Paths, Application Guides, and More Things | 1:51:39

All right.

Welcome to ActiveInference Stream 10.2.

It's February 23rd, 2025.

I hope we will stay online long enough through the storm for what should be a very interesting time.

So thank you everyone for watching.

I'll keep an eye out in the live chat.

Let's jump right into it.

All right.

I'm going to start with a GitHub push and an Obsidian push.

So this is all going to be going to the repo ActiveInference Institute slash cognitive.

Starting the stream.

GitHub push.

Then.

Over in Obsidian.

I'm going to do an Obsidian push.

So that will be going live to the publish.obsidian.md slash active inference page.

And we will.

Select the changed documents.

And the new 138 documents.

Okay.

Obsidian push going live.

GitHub push.

Double check it.

And then let's jump right in.

All right.

We are in it.

So.

This is going to be a little bit wild.

There's parts that are in various and multiple states of functionality.

It's just too fun not to share though.

So I wanted to update and follow up on the plain text series.

And continue to give people some checkpoints and milestones.

And open the space for their suggestions and contributions to some very exciting new ways of learning and applying active inference.

So.

Let's start with the agenda.

Here's the windows that I have for reference.

I'm going to put the GitHub window away.

Maybe we'll push later in the stream.

There's going to be three main windows up.

There's going to be a browser where we have the GitHub repo and the Obsidian publish format.

And some other stuff is going to pop up in here later.

We have the Obsidian window here where we can do force directed graphs.

And this was used also in the previous plain text stream.

Then we have cursor 45.

Running along.

Mostly using agent mode with the Claude 3.5 sonnet.

So.

The stream.

Being the second in the plain text series.

Is going to look at learning paths.

Application guides.

And more things.

The stream will focus on advancing our understanding.

And application of.

Active inference and cognitive modeling concepts through.

Learning paths.

Detailed application guides.

Initially for rxinfer.jl Julia package.

And a generic things repo with a test suite.

That's almost too comprehensive.

Giving some challenges still.

And other resources.

We're on YouTube and elsewhere.

The information is all linked here.

And we'll see where CodeViz comes into play.

Alright.

We began with the GitHub push.

Let's check in first with the generic things.

Okay.

The visualizations that were only minutes prior.

Working just fine.

Of course failed to work.
But that's totally fine.
They really are all in there.
And I'm going to pop over to.
The things.
Generic thing.
This is.
Seeking.
And several.
Dozens of prompts.
And scribbles.
Along the way.
Towards a foundational.
Implementation of active inference principles.
Through a generic.
Quote.
Thing.
That can engage.
In message passing.
Belief updating.
Free energy.
Minimization.
In a.
Potentially interactive.
Nested.
Federated situation.
It's a little bit of a balance.
With the markdown files.
Because there's things that look good.
On the markdown display.
Of GitHub.
And less so.
In.
Obsidian.
And vice versa.
So let's look over to it.
In.
Obsidian.

The.
Core.
Idea here.
As has been.
Suggested.
And.
Indexed.
Towards.
By many.
Over the years.
Is.
What is that.
Generic.
Active inference thing.
What is it.
That we can have.
That's.
Such a flexible.
Schema.
That it'll allow us.
To deal.
In.
Continuous.
And discrete.
Time.
Continuous.
Discrete.
State.
Spaces.
Single agent.
Interacting.
With the environment.
Multiple agents.
Dissolving agent.
Environment.
All these different kinds of ideas.
That are out there.
And then.

Where plain text.

Is heading.

And this repo.

And this.

Vector.

Of collaborative research.

Is.

With a lot of the.

Pieces of information.

Be they.

Analytically.

Former.

Formal.

Expressions.

Be they.

In.

Even.

Probably.

Evocative.

Other.

Plain text.

Representations.

And eventually.

Multimedia.

Then.

When we say.

Implement.

Variational.

Free energy.

Minimization.

Or implement.

Expected.

Free energy.

Minimization.

We can.

Know.

That it's going to use.

Retrieval.

Augmented.

Generation.

RAG.

Like.

Approaches.

To.

Use.

That information.

In the repo.

To structure.

The discussion.

So.

A lot of the.

Focus.

Over the last.

Several days.

Let's.

Resume.

It's.

Just in terms of.

What cursor is doing.

In the background.

I'm having it.

Run the test suite.

Again.

And again.

While.

Let's see.

What these ones.

Will not.

Look.

Like.

Having it.

I'm having it.

Run the test suite.

In the background.

In agent mode.

Where it's going to.

Make changes.

Rerun the test suite.

And so.

I found that this kind of.

Agent.

Semi-autonomous.

Semi-supervised.

Test suite.

Operation.

Is super effective.

Um.

But here's a few.

Of the key.

Elements.

So.

I wanted to have.

The capacity.

To.

Have the generic thing.

Of course.

Interact with other things.

This could be.

Alice and Bob.

Just.

Having that.

Classic.

Conversation.

That quantum.

Classic.

Conversation.

Or it could be.

Agent environment.

Or it could be.

Multi-agent situations.

Here's the.

Concurrency.

Diagram.

In terms of.

The message passing.

Now.

This was a very.

Interesting topic.

That we'll probably.

Return to.

When we get to.

Rx inferred later.

Is.

To what extent.

Do we want to.

Schedule.

The logistics.

Of message passing.

Because.

As it is.

With all this.

General.

Active inference.

Thing.

Research.

It's like.

What is the least.

Opinionated schema.

Nothing.

What is the most.

Opinionated schema.

Well.

It might be super effective.

For where it's.

Effectively.

Opinionated.

But.

It's not going to have.

Degrees of freedom.

So.

Which are the.

Degrees of freedom.

That we really.
Want to engage with.
When.
For example.
Reactive message passing.
In Rx infer.
Which we're going to.
Return to.
Makes some of this.
Message logistics.
Possibly.
Not necessary.
For the scheme.
But again.
If it's.
A scheme.
That's.
Maximally.
Unopinionated.
It's just.
An empty.
Repo.
I.

I.
I.
Focused.
On the.
Message passing.
Functionality.
Where.
Starting with.
Three.
Things.
That were.
Connected.
In a.

Linear.

Chain.

With.

A lot.

Of.

Test.

Functionality.

Within.

The message.

Passing.

Architecture.

Hello.

Jeff.

And.

Monir.

Here.

Is.

The.

Logistics.

Of.

The.

Action.

Perception.

Loop.

Inactive.

Inference.

Again.

One.

Could.

Remain.

Even.

More.

Unopinionated.

On.

The.

Ordering.

Of.

The.

Action.

Perception.

Loop.

The.

Ordering.

Of.

The.

OODA.

Loop.

And.

That's.

A.

Degree.

Of.

Freedom.

That.

Rx.

Infer.

Allows.

Us.

To.

Relax.

But.

Here.

I.

Wanted.

A.

Language.

Independence.

Scheme.

And.

Just.

A.

Draft.

Hopefully.

One.

Where.

We.

May.
Actually.
Come.
To.
Realize.
That.
We.
Do.
Want.
To.
Make.
The.
Schema.
Already.
Pre-adapted.
To.
Reactive.
Message.
Passing.
However.
If.
We.
Relax.
The.
Logistics.
Here.
So.
That.
The.
Reactive.
Message.
Passing.
Can.
Proceed.
With.
The.
Least.
Opinionation.

Then.

That.

Schema.

Wouldn't.

Be.

Able.

To.

Be.

Implemented.

Outside.

Of.

A.

Reactive.

Message.

Passing.

Scheme.

Because.

It.

Be.

Like.

Well.

Where.

Are.

The.

Logistics.

And.

Even.

Within.

A.

Reactive.

Message.

Passing.

Scheme.

We.

Still.

Do.

Need.

To.

Have.

Certain.

Kinds.

Of.

Event.

Based.

Injections.

In.

Order.

To.

Get.

The.

Reactivity.

Going.

So.

We.

Have.

Environment.

Sensory.

Internal.

And.

Action.

States.

External.

Internal.

Sense.

And.

Action.

Being.

The.

Blanket.

States.

Environment.

Passes.

Observations.

To.

Sensory.

Which.

Passes.

A.

Belief.

Updating.

Message.

Okay.

Here's the test suite.

Being emitted.

So now.

This is what.

The autonomous test suite.

Improvement looks like in cursor.

It pops out.

The test suite.

It.

Is passing many of the tests.

But some of them are still failing.

And it makes a HTML.

Website.

That has a lot of useful information.

A massive.

Fractal.

Test suite.

In fact.

And then.

Just continuing to say.

All right.

Fix it.

Methodically.

Agentically.

Professionally.

Comprehensively.

Continue to run the test suite.

And go on.

And do what you can.

To fix it.

And so on.

Those kinds of encouraging.

Fractal.

Prompts.

So that for each of these core.

Functional.

Elements.

Of the generic thing.

Concept.

The core.

Which we'll look at.

Free energy.

Processes.

The inference process.

Markov blanket.

Depiction.

The message passing.

Which could be.

Just the standard.

Kind of.

Message delivery.

Pub sub type.

Or it could be more federated.

Set up.

Being needed.

And a visualization.

Script.

Here.

Then we have.

The test suite.

In the tests folder.

Which.

Gives.

These kinds of outputs.

So it's going to continue.

Debugging.

While we.

Look through.

From the internal states.

Expected free energy.

G.

Across the pi.

List of possible policies.

Is calculated.

Then.

Action is selected.

From.

The policy.

Posterior.

And then.

That.

Executed.

Selection.

Of action.

The one hot action.

Gets passed back.

To the environment.

So.

There are other.

Topologies.

And other ways.

To schedule.

The logistics.

Of this message passing.

But again.

To the extent.

That we're less opinionated.

About some of this stuff.

We actually have like.

Less and less.

Of a schema.

So.

Without.

Just.

Having a piece of blank paper.

Or an empty repo.

Empty mind.

In front of us.

And say.

Well this is generic.

Active inference.

How do we put something.

Into that.

Void.

And.

More information.

As with much.

Of.

Procedurally.

And probabilistically.

Generated code.

It has all kinds of.

Quirks and features.

So.

It isn't high reliability.

Code per se.

But it's awesome.

For what it is.

And.

It's a starting point.

For.

Line by line.

Manual.

Investigation.

And curation.

And also.

For.

Further.

Contemporary.

And future.

Advanced.

Code repair.

Strategies.

So.

Let's.

Let it continue.

Doing the test suite.

It's not.

Giving the visualization.

It makes several dozen.

Visualizations.

I hope that we can get there.

By the end of the stream.

Somehow.

Some way.

With several dozen.

I know that there's like a kind of.

See it's a believe it element.

And certainly.

The visualizations.

Provide.

Really important.

Visual.

Calibration.

And gut checks.

And accessibility.

For these processes.

That are happening.

So we'll check back in.

We'll make sure that the.

Autonomous.

Test suite.

It'll stop.

After 25.

Agent tool uses.

I don't think that's.

Needs to be that case.

But it is.

The case.

In.

This version 45.

With this current agent.

But just so we remember.

Ensure that.

If and where possible.

We'll let it continue.

What it's doing now.

But then it'll eventually say.

Okay.

That's all I could do right now.

That.

Visualizations.

Are made.

And.

Prioritized.

And.

Fully.

Logged.

With.

Full.

File.

Name.

To.

Terminal.

Okay.

So.

We will.

Return to the generic thing.

That's why.

Starting to debug.

The generic thing.

Was first on the agenda.

All right.

Now to.

Learning paths.

So.

This is.

In.

Okay.

Just output another.

Test report.

This is in.

Docs.

Guides.

Learning paths.

Okay.

So.

In the prior.

Plain text.

10.1 stream.

We were having.

A ton of fun.

Let's look at it.

In both.

GitHub.

I mean.

We have three ways.

Of looking at this.

We have the.

Cursor.

Local.

So.

This is the code editor.

Version.

Then.

Looking also.

Locally.

Is.

Obsidian.

And then.

Emulating.

That file.

Structure.

Of the repo.

Is.

The.

GitHub.

But that won't be updated.

Until.

The.

The push is made.

But we can.

We can make that push.

Just for fun.

Let's see if there was any visualizations made.

Okay.

Okay.

Okay.

Okay.

Thing.

Generic thing.

Visualization.

Let's just look at where it gets.

So.

Thing.

Generic thing.

Visualizations.

Okay.

Here we go.

Just to.

Sort of confirm that the visualizations are happening.

In the debugging.

So here we go.

This is the.

Action.

Repertoire.

With summary.

As well as.

Precisions on actions.

And some eigenvalue spectra on that.

This is the.

Action.

Repertoire.

These are belief states.

And uncertainty.

These are output.
As part of the test suite.
So that's why these.
Have.
Each time a test suite is run.
Some of the.
Time stamping differs.
Other.
Outputs are just.
Without time stamps.
So they're overwritten.
Here's belief updating.
Not sure how.
Accurately.
It's implementing.
Bayesian updating.
At least though.
The skeleton.
And the architecture is there.
Free energy.
Histories.
Here's.
Here's the.
Free energy landscape.
From a parameter sweep.
Part of the free energy.
Minimization test.
Looks like it's the same.
Across tests.
Here's the.
Hierarchy.
Some.
Funky stuff happening.
With the labeling.
But.
A parent.
Has.

Child.

Child.

Child.

Child.

Child.

Child.

Child.

And then a grandchild.

Here are the.

Sensor.

To processor.

So that's.

From.

Sensory.

To internal.

And then from internal.

To action.

Patterns.

Of input.

And output.

It.

Here's.

The triadic.

Internal.

And blanket states.

For the thing.

This is a two.

Thing.

Message network.

Processor.

See.

Now it's.

Adding more files.

As this next.

Round of the test suite.

Continues to occur.

More.

Belief state.

Distributions.

Correlation heat maps.

Summary statistics.

Eigen values.

On.

Matrices.

Some very.

Interesting.

Possibilities.

And hello.

Ender.

For.

Spectral analysis.

On these random.

Matrices.

Temporal evolution.

This is a little bit of a.

Kind of.

Pseudo.

Mondrian.

Asking it for full.

Details.

Give the full.

Details.

The whole.

Repertoire.

Of the sensory.

The internal.

Only action states.

And those can be.

Described.

As.

A through time process.

As a.

In time.

Process.

Temporal analysis.

Looking at.

Sensor.

Processor.

And actuator.

These three rows.

And.

The eigenvalue.

Evolution.

Test.

Belief updating.

Okay.

Another Bayesian.

Free energy landscape.

This is a nice looking.

Free energy landscape.

And.

Roughly.

Concording.

With what we would.

Hope to see.

Which is.

On this.

Manifold.

Where the internal.

States.

Like negative two.

And negative two.

For internal.

External states.

Where they track each other.

They have.

A lower.

Free energy.

And to the extent.

That they're off.

That.

Manifold.

The free energy.
Is higher.
And.
A nice.
Tetrahedral.
Double zigzag.
Message.
Passing network.
Okay.
So.
We're going to.
Let the.
Autonomous debugging.
Proceed.
While we look at.
The learning paths.
These learning paths.
Are for.
People.
And.
For.
Non-people.
So.
The way this worked.
Is again.
Pop back.
However many minutes.
To.
Before.
Where.
We saw from GitHub.
That the visualizations.
Were being output.
We had all these.
Different.
Topics.
Across the knowledge base.
Which were more like.

Idea topics.

Just to kind of.

Remind on this.

In the knowledge base.

There are.

These.

Folders.

So.

Augmented reality.

And the spatial web.

And.

When you are in.

Obsidian.

The node.

That you're in.

Sometimes it's a little hard to find.

In a big network.

So here's spatial web.

And then.

The.

What it's linked to.

Are.

Topologically.

Connected.

And we can drag it around.

And see what.

Comes along for the ride.

So there's like.

That spatial web.

And then connected to.

Active inference.

So these are.

Topics.

And these.

These.

These are like.

Fractal.

Wiki.

Pages.

They could be.

Line by line.

Line.

Versioned.

And annotated.

And calibrated.

As.

As context windows in LLMs increase and as the efficacy of putting very large knowledge corpuses into finite context windows also improves, relevance realization improves, there's really no downside to adding more.

Including a plurality of perspectives on formalisms, on qualitative points, these are all good and exciting things.

So in the knowledge base folder, I've added a few dozens of more topics since the last stream.

So some of these are, let's see, they look like they're, okay, it just sometimes takes a little bit old Linux laptop doing what it can.

Connecting with a lot of Bucky Fuller stuff, which those going through the back catalogs will know has long been a possible, promising, exciting connection with synergetics and tensegrity and active inference.

But these are knowledge base files.

Here's the mathematics folder.

Folder.

So calculus.

This could be just one very, very long, several hundred page textbook.

It could be many textbooks.

It could be calculus one or it could be calculus adding more adjectives and so on.

Category theory, same thing.

Complex systems.

And also, it can get really specific, like de-separation, which is a criterion for determining conditional independence relationships in Bayesian networks.

So going around and learning through each of these topics is like exploring and super fun.

And yes, needing to be verified and checked and always possible to add, but it's like exploring in a lightly synthetic, intelligently peer-reviewed network.

In the cognitive folder, the approach that I took was to make a file called cognitive phenomena.

And in Obsidian, you can choose, first off, what you want the highlight color to be.

So in the appearance, you can have accent color and it can be any color you like.

So I like red.

The files that are linked and exist are bright red.

And then there's the dimmed ones, which don't exist.

But a single click will make that file and just drop it into the top level of the repo.

So the approach I took with a cognitive folder was to several times iteratively prompt.

Okay, let's continue with the test suite.

No, we default stop the agent after 25 tool calls.

Please ask the agent to continue manually.

Please continue to run and fix and improve the test suite.

Okay.

So several times iteratively, agentially, methodically, comprehensively, with adding a little bit of spice from the sideline, throwing in fly balls from left field and all that, added in different topics to this cognitive phenomena index file.

And then said, now, comprehensively, make the files that don't yet exist.

And so this can be iteratively done, kind of like a two-stroke meta engine where build out the repo index.

Add more cognitive phenomena.

And there's no end to how much specific detail could tessellate down.

And then the other stroke of the engine is, for the articles that don't yet exist, make them.

And for the extant articles, improve their dense obsidian-style linking and improve their technical and analytical and applicability and accessibility and so on.

One note, I'll just footnote.

This is all in English language, but there's no reason why it couldn't be en masse translated in Snapchat.

So that was the cognitive folder.

Citations.

This one is empty.

Really exciting collaborations with the Gen25 team and others.

So I look forward to including people and documents into this as well.

But for now, there's no people and there's no papers.

Then, over the last little bit, I also added in a lot of entomology and apidology, which is B studies.

Related topics.

And myrmecology, which is ant studies.

So then, later on, as may very well happen, when we say something like,
make a generic thing specialized for applying to collective intelligence in ants.

That is very well primed and structured for the LLM to be able to make that adjacency and bridge
between what we already have about ants and what we already have about generic things.

So this is in the, there's also some fun topics like mathematical entomology, which I don't really even know if
I've heard of it done specifically that way.

But it sounds like it'd be a really exciting class or topic.

Kind of connecting population biology, evolution, ecology, development, spatial modeling, cognitive science
and behavior, experimental design,

applications like for integrated pest management and bioregional stewardship and all these other things.

And then, it's like, here are five current research applications for mathematical entomology.

People can have different perspectives on where or how they would value these research applications
or how close they are from the bench to the bedside or whatever the ENTO equivalent is,
but it's like something that can be added to.

And then, also, the knowledge base can include specific things,
like here where we've done a lot of discussion on the biofirm.

And this is like a documentation fractal about the biofirm addressing the kinds of topics and questions
that anyone exploring the biofirm, whether they be inner developer, membrane communicator,
or external, somehow relevant to learn across this.

It's like, how is active inference applied in the biofirm?

Well, here's an interactive article on that.

And this is all available online.

Like, here's what it looks like online.

Here's how the biofirm is doing multi-scale nested ecological active inference

and applying the free energy principle.

And then, lines can be corrected or improved or added.

But again, to get from a first pass to here so rapidly is so fun.

Okay, so that's the knowledge base.

Now, back to the learning paths.

This was a new piece over the last few days slash weeks.

So here, um...

Still debugging away.

Go get it, cursor.

Learning paths.

Again, over the years.

So many discussions.

Well, what's the best entry point?

And it was sort of like, the truth is a pathless land, like Krishnamurti angle.

You know, what is the one entry point?

And sort of a recognition that there isn't just one.

There's not just one entry point to anything.

And so now, the interactivity of these tools is actually helping shorten the active inference cycle and time, OODA loop timing,

to an absurd level for live, custom, generative, educational material generation, which could be done one shot.

Like, you could just go to your favorite LLM, generative AI, chat interface, and say, I like X.

Please develop me a curriculum for learning active inference.

So that's something that didn't exist five years ago, or ten years ago, earlier on in learning active inference.

But that's something where anyone with access to those chat interfaces can get that one shot.

But this is opening up with the plain text repo,

the possibility for having these co-constructed and legible and credible of different kinds learning paths.

And one person asked in the Discord,

so the learning path, I forget which way they even asked.

Like, does it go from active inference to the area or from the area?

It's like, well, paths go both ways.

So how about active inference?

Okay, I'm just, I think it just dropped for a second.

Okay, okay.

It should be back.

But let me know if anything in the chat, if anything weird happens.

Too many weird things are happening to even say.

But I think you all know what I mean by that.

So this active inference economics intersection
can be used to take someone from a more generic active inference place
or to the domain,
or it can be run in the other way.

I'm going to just drop the bitrate on the stream a little bit.

Right on.

So, this is what it looks like in the local Obsidian.

So here we are.

And this connects.

Here's the active inference economic learning path.

Let's go over to where it is in.

So now we're going to look at that.

I'm going to control A, control K,
and let's use the Grok2 model.

Comprehensively, vastly add relevant sections
and ensure linking style with extant files.

Okay.

So here it is.

Running through.

Adding information.

To these sections.

We'll see if it adds any new sections as well.

Okay.

Ooh.

Didn't like those parts.

I think that in the coming versions of Cursor,
they're going to be improving this
and simplifying their kind of fractal product offering in a way.

Unifying the chat in the Composer.

Or so I heard from Gary.

But we can add this.

It's just that I have it already running in the Composer
with the debugging.

Which maybe we can just sort of...

We kind of get the picture on the generic thing.

We get it that we can write
megafractal modular test suites.

And we can use that
to do this autonomous, driven development
on different topics.
And we'll return to it later in the stream.
So I'm going to start a new chat
with the economics learning paths.
Comprehensively, develop this more.
We want to use this for a convening
that leads to a three-quarter
co-learning course
between an economics department
and the Active Inference Institute
so that all backgrounds and familiarities
with e.g. math, biology, economics, etc.
are included
and have many contribution avenues.
So then it's going to be
doing this modification
in Cursor
that'll be live represented
in my local Obsidian.
And then we can push to GitHub
for it to then...
And do the shift publish
to have it also go live here.
So then the idea of these learning paths,
one idea of them,
is biological learning path.
Like this URL,
or something very much like this URL,
is like a stable URL.
And it can continue to be developed out.
So instead of where we would be
just a couple months ago,
okay, I'm kind of curious about this.
I really like to do pottery
and embroidery and drawing.
And this is my favorite.

You know, it's just like,
where do you start with Active Inference?
Oh, well, here's the 2022 textbook.
And here's a textbook group.
That's awesome.
That's really, hopefully, exciting.
That's not every single last mile,
let alone every last themtometer.
Now, there can be
Active Inference Mermecology Learning Path.
So that's from somebody who's studying
the generalized Active Inference topic,
that can take them out
to the Mermecological last miles.
And from somebody who is working on ants in the field,
take them in towards,
this is just one way of studying it.
Okay, here's, it just laid it out.
And we'll say,
yes,
make all these additions
to the economic learning.
Hello, reconfigurability trainer.
Yeah, anyone,
if you have like a suggestion
for another learning path
or a fun thing to try.
Basically, we're in the middle of act two
of this stream
where we're going to look at the learning paths,
develop the learning paths a little bit.
Then act three
is going to go over to RxInfer
and explore some para documentation,
some fanfic documentation that I made.
And then part four
will be wherever the live chat goes.
Okay, so here

is the modifications
that cursor is making
to this economics course.

So again,
just this is kind of a micro comment,
but when we use the control K mode,
it deleted large sections.
And then here,
it even recognizes
through the iterative process,
I noticed some examples were deleted.
I'm going to add them back in.

Okay, so that's happening.

So now this
has been expanded
into that multi-quarter
course
concepts.

Okay,
so that's an example
of improving a learning path.

It could be fractal
unrolled
into any level of depth.

It could be,
we have 15 minutes
and it's a ninth grade classroom
in this language
and here's what they're most excited about.

Or it could be,
it's a whole lab
and there are five PhD students
who are ready to spend five years
on this topic.

Okay.

Munir has made a suggestion.

This is going to be
active inference

observation

learning

path

dot md.

So,

Munir wrote,

human judgment

and active inference

especially

the observer effect.

Okay,

I'm not sure

if you meant this

to be a learning path

but I read it.

So,

oh,

okay.

And then

followed up with,

this is just a suggestion

to see what active inference

agent's self-perception

about its own

belief model.

Okay.

Let's turn this into

please comprehensively

continue to

develop

these ideas

into a learning path.

It's in the folder

with all the other

learning paths

so it knows

a little bit

about the structure

and the template

and all of that

as a

two-way

street

beep beep

for

those

interested

in

such

topics

like

as described

and

observation

the art

of observation

the magic

and possibility

of

observation

and active

inference

parentheses

agents

and

link

obsidian

style

accurately.

anyone

can

clone

the repo

pull

the

active

inference

institute

slash

cognitive

repo

get it

going

locally

navigate

over to

this folder

make a

new

file

we'll do

a

mirror

macology

learning

path

too.

okay

but

here

we're

on this

observation

learning

path

so it's

like

kind of

like

phenomenology

kind of

like

ethology

this

learning
path
explores
the
profound
relationship
between
observation
human
judgment
and
active
inference
it
examines
how
the
active
observation
itself
shapes
our
understanding
and beliefs
while
connecting
these
insights
to
the
mathematical
framework
of
active
inference
the
path
integrates

perspectives
from
cognitive
science
philosophy
of
mind
and
computational
neuroscience
to build a
comprehensive
understanding
of how
agents
both human
and artificial
perceive
learn from
and influence
their environment
through observation
let's
look over
at it
in the
obsidian
setting
so here's
the observation
learning
path
okay
here it
is
little red
dot
here's

what comes
along
for the
ride
when we
force
drag
it
so
let's
see
foundations
of
observation
phenomenological
approaches
and
scientific
interesting
inner
and outer
observation
from the
blankets
perspective
oh wait
active
inference
basics
pre-energy
principle
predictive
processing
core
concepts
observer
effect
quantum

measurement
theory
social
observer
effects
active
inference
and self
perception
belief
updating
and interoception
the art
of observation
mindfulness
and attention
advanced
observation
techniques
oh brother
pattern
recognition
detail
perception
contextual
awareness
advanced
topics
computational
models
applications
and
implications
projects
and exercises
resources
essential
reading

the
mathematics
of mind
active
inference
and agency
by Carl
Friston
I don't
know if
that's a
real paper
but it's
like we're
at the
point where
it's like
does it
matter
because we
could generate
what would
that be
okay
would you
like to
add specific
content
expand the
resources
yes
vastly
expand the
sections
related to
exercises
there
should

be so
many
accessible
tangible
everyday
experiences
that will
relate to
people's
direct
sensory
and
cognitive
experiences
it should
be a
vast
list
and one
with
many
exciting
starting
places
for
inner
outer
and social
exploration
meanwhile
let's just
get the
mirror
mycology
one
write a
comprehensive
learning

path
document
for
mirror
mycology
we are
a
team
of
enthusiastic
quote
field
researchers
we are
interested
in
collective
intelligence
parenthesis
and also
via
okay
we'll look at
that in obsidian
in a second
and also via
nested active
inference
interested
in
the
nest
mate
or the
colony
make
the
learning

path
now
please
okay
new
chat
all
right
so now
while the
mirror
mycology
path
is
unfolding
let's
look at
the
observation
one
thank you
Muneer
for the
suggestion
that was
awesome
okay
now we're
looking back
to the
observation
all right
here are the
projects and
exercises
let's see if
any of these
strike our

immediate
fancy
inner world
exploration
sensory awareness
exercises
one
body scanning
practice
i know of
one cat
that's very
interested in
that
two
breath
observation
series
three
emotional
weather
tracking
second
section
of
inner
world
exploration
cognitive
observation
exercises
thought
pattern
recognition
memory
and perception
studies
attention

direction
experiments
wow
outer
world
exploration
part one
environmental
observation
projects
nature
observation
series
urban
environment
studies
object
deep
dive
series
whoa
single
object
extended
study
material
property
investigation
function
form
relationship
object
lifestyle
tracking
kind of a
nod towards
systems
engineering

pattern

recognition

exercises

visual

pattern

detection

sound

pattern

analysis

time

pattern

studies

social

world

exploration

interpersonal

observation

projects

nonverbal

communication

study

conversation

pattern

analysis

group

dynamics

observation

cultural

pattern

recognition

ritual

routine

studies

media

pattern

analysis

and

computational

analyses

okay

now we

return to

mirror

ecology

yes

make the

learning

path

much

longer

and

include

e.g.

math

diagrams

artistic

experience

possibilities

okay

while that's

improving a

mirror

ecology

path

let me

check

the live

chat

okay

reconfigurability

wrote

what are

use cases

that this

can apply

to

what do
you think
looking forward
to your
thoughts
Munir
thank you
again
Andrew
still
catching up
from start
at 2x
speed
but
you're
you're
having the
1x
experience
of the
2x
speed
the economic
and biological
learning paths
are very
illustrative
examples
what about
social science
curious how
more specific
and general
impacts
and outcomes
yeah
I mean

one answer
of course
is I would
look forward
to you
adding that
file and
the github
repo
making that
social sciences
or we can
make a stub
here
okay
let's
let's
yes
add
many
ant
related
topics
colon
deep
lore
art
culture
and folk
ways
amazing
historical
and
accurate
natural
history
ant
anecdotes

then we'll
look at it
in obsidian
okay
reconfigurability
has the topic
of underwriting
been explored
I know
that you really
like that
so let's
do a
learning path
active
inference
underwriting
learning
path
dot md
this is a
comprehensive
make this
a
comprehensive
learning
path
to and
from the
concept of
underwriting
give
special
thought and
attention
to the
domain
specific

ontologies
that
beginners
and
experts
across the
active
inference
and
underwriting
domains
may
bring
complexly
to such
an
intersection
okay
looks good
for the ants
we'll come back
there
bring
complexly
to such
an
intersection
please
be
vastly
comprehensive
about
ontology
narrative
tools
etc
for
underwriting

structure
this
massively
okay
make
this
happen
agentically
comma
methodically
okay
now while it's
doing the
underwriting
learning path
let's return
to our
mimicology
okay
so it's
off
no
this
sometimes
happens
it's off
by itself
all by
its
lonesome
but
here's
how you
edit it
in obsidian
double
brackets
and then

start
typing
first
it
suggests
it
to be
there we
go
it just
took a
little
bit
now
we
linked
it
to
mirror
mycology
so
now
it's
a red
link
because
we
already
have
that
file
mirror
mycology
so
now
here
we
are

now
here's
mirror
mycology
connected
to
earth
system
schema
interesting
okay
so
let's
see
active
inference
mirror
mycology
learning
path
the
learning
path
integrates
mirror
mycology
the
study
of
ants
with
active
inference
and
collective
intelligence
frameworks
it is

designed
for
researchers
interested
in
understanding
multi-scale
cognitive
processes
in
ant
colonies
from
individual
ants
to
colony
level
behavior
so
this
is
the
crux
of
the
issue
later
models
and
other
knowledge
niches
will
not
fall
into
such

a
trap
it's
all
about
what
you
call
the
individual
ants
presupposing
it as
nest
mate
but
it's
like
okay
I'm
somewhat
startled
I'm
surprised
from
nest
mates
to
colony
level
behaviors
of
ants
incorporating
mathematical
computational
and artistic
perspectives

so
math
computational
core
concepts
of
mirror
ecology
active
inference
and
social
insects
research
methods
field
laboratory
advanced
multi
scale
the
individual
level
it's
all
good
it's
all
good
nest
mate
level
single
nest
mate
but
now
this

learning
path
can be
used
in
generating
other
materials
and it
influences
and so
even though
it's an
ontolinguistic
perspective
that I
have
that
haphazard
usage of
the term
individual
to refer
to
the
nest
mate
which
would
be
equally
inappropriate
if it were
haphazardly
used to
describe a
colony
is

what makes
the
colonies
behavior
seen as
collective
intelligence
implicitly
supposing
that the
nest
mate
is the
individual
whereas
of course
the
individual
quote
nest
mate
or just
the
nest
mate
is
as with
all
cognitive
systems
its own
composition
artistic
creative
dimensions
let's
let's
have

the
yes
please
vastly
continue
to
add
relevant
topics
sections
exercises
trainings
reconfigurabilities
to
this
underwriting
document
underwriting
learning
path
documents
okay
mermiculture
art
scientific
illustration
technical
drawing
ant
morphology
colony
architecture
behavioral
sequence
sketching
wow
this is
very

classic
environmental
context
drawing
but if
you look
back in
Holdo
Blue
and
Wilson
and all
that
if not
much
much
earlier
this is
what
was
there
and
some
of
those
are
still
in
use
and
now
with
generative
AI
we
could
make
it

look
like
it
was
a
different
species
and so
on
so
kind
of
cool
digital
art
creative
documentation
photography
and
videography
wow
that
reminds
me
documentation
three
art
science
integration
installation
art
public
engagement
wow
diagram
types
to
master

scientific
diagrams
SD
mathematical
visualizations
artistic
representations
and then
we'll
just
add in
category
theory
extended
practical
projects
awesome
okay
mathematical
appendix
readings
interesting
interesting
here
the
super
organism
by
hold
open
wilson
now
this
is
like
a
situation
where

the
title
of
the
book
falls
into
the
trap
well
if the
colony
is the
super
organism
then
that
must
mean
that
the
nest
is the
individual
organism
and that
is the
concept
that has
grasped
the
imaginations
and
constrained
of many
wildly
interestingly
historically

active inference

by par

pazulo

friston

et al

but it's

like here's

where the

line wise

editing

matters

par

pazulo

!

friston

2022

MIT

press

there

is

also

a

textbook

group

about

this

book

at

the

active

inference

it'll

take a

second

to load

this

at

the

active
inference
institute
tools
and
equipment
additional
dynamical
systems
in
social
insect
even
the
concept
of
calling
the
study
of
ants
and
bees
social
insects
is
like
oh
so
then
the
non-social
would be
the
nest
mate
and
then

the
nest
mates
are
hanging
out
and
that's
the
composition
of
the
social
so
it
is
baked
in
so
deeply
into
the
language
of
what
is
social
and
solitary
insect
what
is
individual
and
collective
intelligence
wow
and now

we can
look at
it
line
by
line
career
paths
fun
okay
oh
next
steps
join
the
local
myocological
society
everywhere
could have
one
all right
let's go
to the
underwriting
all right
then this
will be
the end
of this
second
phase
then we'll
look at
rx infer
documentation
para
documentation

fanfic
and then
we will
go to
phase four
of the
stream
with just
whatever
else
people have
added in
the live
chat
so thank
you
reconfigurability
for this
suggestion
and right
now
intra
stream
update
so i'm
going to
push that
to github
then in
obsidian
shift
period
i set
that
you can
set your
own
shortcut

changed
and
new
33 new
publish
so that
means that
the
underwriting
what's
a
myrmological
community
yeah
reconfigurability
i don't
know
but
myrmecology
is the
study of
ants
so the
myrmecological
societies
would be
like local
ant
enthusiast
researcher
learner
chapters
okay
but let's
look at it
so now
we've pushed
it to

obsidian
publish
so let's
just look
at how
it looks
in
docs
guides
learning
paths
here's
underwriting
and then
now over
here
here's
underwriting
so i'm
going to
put this
link in
the chat
reconfigurability
there you
go
all right
this
learning path
provides a
comprehensive
guide to
understanding
the intersection
of active
inference
and underwriting
designed for

practitioners from
both domains
seeking to
bridge these
fields
some of whom
might be in
the live
chat i don't
know
prerequisites
and this is
where it
could link
again
more densely
i'm going to
have it do
this in
cursor
comprehensively
link
obsidian
style
to
relevant
file names
for
e.g.
prerequisites
and
everywhere
possible
so
prerequisites
probability
theory
risk assessment

basic machine

learning

statistical

modeling

you know

again

somebody

could say

well

i want

there to

be a

in the

in the

phase space

of math

and computer

science

all the

quadrants

can be

explored

it could

be a

no math

no computer

science

version

it could

be

math

but no

computers

index

card

style

could be

computers

and no
math
gen z
style
no i'm
just kidding
it could
be
math and
computer
science
together
um
okay
domain
the underwriting
domain
ontology
this is
where
the
domain
experts
can say
okay
what else
needs to
be added
here
in terms
of the
core
concepts
key
processes
act
inf
domain

ontology
this
draws
on
the
models
training
which
is
probably
also
on
earlier
versions
of the
act
inf
ontology
but also
um
the
files
okay
now it
has
upgraded
all of
these
so now
this is
cool
so it
upgraded
all of
these
into the
obsidian
linked

format
so save
that
now in
obsidian
i'm going
to do
shift
period
changed
here's where
the underwriting
learning path
was changed
publish it
now
wow
so cool
incredible
offering
by obsidian
their software
and their
sync
feature
it
happened
live
didn't
it
so now
we can
look at
it in
our
local
setup
here's

now what
gets
dragged
but
previously
it was
not
connected
to
anything
but
now
all of
these
and then
these
ones
that
are
dimmed
these
ones
could
be
added
but
now
oh
probability
theory
oh
here's
what
probability
theory
is
oh
what's

a
sigma
algebra
oh
well
you know
that
could
be
linked
but
back
to
the
underwriting
key
processes
each
of
these
could
be
documented
and
then
fundamental
concepts
for
active
inference
like
yeah
what
is
variational
inference
open
that

up
in
a
new
tab
oh
it's
a
method
for
approximating
complex
probability
distributions
through
optimization
okay
okay
intersection
integration
bridging
concept
risk
modeling
decision
making
and
learning
systems
application
areas
automated
underwriting
portfolio
management
claims
processing
tools

and
technologies
technical
stack
okay
learning
progression
level
one
understanding
basic
concepts
so
underwriting
principles
again
this could
be
it says
the link
underwriting
principles
does not
exist
but now
in here
in the
relevant
location
in
knowledge
base
add
and link
to and
from here
the
quote

underwriting
principles
file
make it
accurate
comprehensive
applicable
and more
okay
so
understanding
basic
concepts
of
underwriting
active
inference
and risk
assessment
tool
familiarizations
integration
advanced
implementation
we could
think
oh
what about
multi-agent
modeling
in this
situation
just
add it
foundations
of risk
assessment
now it's

getting in
another level
of detail
active
inference
fundamentals
practical
implementations
okay
life
insurance
underwriting
property
insurance
health
insurance
oh
wow
advanced
topics
system
integration
hands-on
risk
calculator
policy
pricer
automated
underwriting
system
four to
six weeks
we could
do it in
four to
six hours
portfolio
optimizer

that one's
tough
three to
four weeks
sounds good
certification
path
level one
level two
level three
industry
integration
internship
program
junior
underwriter
technical
specialist
mentorship
program
continuous
learning
resources
okay
now
here's the
underwriting
principles
so
now
I'll
push it
in
obsidian
here it
is in
the new
guides

concepts
underwriting
principles
okay
so it was
pushed
now
remember
earlier
it said
that page
doesn't
exist
and it
didn't
but now
it's going
to be
see where
it mentioned
it
underwriting
principles
now it's
in purple
so now
here are the
underwriting
principles
utmost good
faith
insurable
interest
indemnity
proximate
cause and
risk pooling
wow

Uberima
Fides
but this
could just
be
semi
manually
or
fully
automated
or anywhere
in between
continue to
unfold
oh risk
assessment
we don't
have that
one
now
compree
intensively
add
and link
the
quote
risk
assessment
file
this will
be the last
part
on this
and then
we'll
move to
rx
infer

and then
anything
else that
people
suggest
okay
but here
we go
risk assessment
it doesn't
exist
okay
now
it's
gonna
write
that
uncertainty
assessment
yeah okay
risk assessment
written
now
even though
it's probably
gonna continue
to update
it
we'll push
on obsidian
so
current state
we click on
that the link
does not
exist
okay
so here's

the old
version of
the underwriting
learning path
now here's
one of the
issues
again
brought up a
little earlier
okay
new
select it
publish it
published
done
okay
so
here
current version
risk assessment
does not
exist
so it's
like
we made
it
in cursor
the prompt
was just
add and link
risk assessment
now
reload the
page
this is
the
on the

I'll
put this
in the
live chat
now
risk
assessment
here's a
whole
article
risk
assessment
is a
systematic
process
of evaluating
potential
risks
and their
impact
in
underwriting
and insurance
contexts
!
this
foundational
concept
bridges
traditional
underwriting
principles
with
modern
active
inference
approaches
to

uncertainty
quantification
and decision
making
it totally
does
wow
so
that
was
fun
thank you
and that
was a
structured
and structuring
educational
journey
as we
explored
a little
bit about
learning
and cognitive
modeling
all right
now
last
section
here
for the
for the
rx
infer
learners
and
practitioners
and then

section

four

would you

use

like

any

other

healthy

whimsical

fun

suggestions

all right

so

here's what

happened

in

the rx

and first

section

okay

all right

well

rx

infer

is

a

software

package

being

developed

by

our

colleagues

and

it

is

updating

and developing

in very
exciting
ways we're
really happy
that we're
partnered with
them and
on this
co-learning
journey
and
recently
they have
done a
big update
on their
documentation
as part
of the
version
4.0
4.01
now
release
and
what's
really cool
about this
documentation
site
other than
the fact
that it
has an
LLM
integration
with
Gemini

so you
can chat
with it
there
but also
this whole
documentation
is a
plain text
file
repo
plain text
repo
in the
Rx
and
Fur
repository
so
this
docs
and this
is a
very
professional
software
codery
way to
do this
these
documents
are
a
website
but also
the source
code
for the

documentation

not just

the code

is also

available

in the

repo

so

what I

did

is I

took

their

SRC

so here's

source

for SRC

right here

from

I did it

within the

last couple

of days

I think

and so

here's

those same

files

so I

just

in

Rx

and Fur

repo

online

I just

downloaded

their whole

documentation
and I
dragged it
in
to docs
then
we're about
to go
through
some
super
fun
you know
we'll see
what names
work
you know
is it
is it
is it
paradox
because it's
off to the
side of the
other docs
is it
heterodox
because it's
different than
their documentation
is it
fanfic
because I'm
just a fan
and learner
and just
looking to
make some

fun
documents
that help
apply
and understand
Rx and Fur
I don't know
but here we
are
let us
go to
readme
so this is
the readme
of Rx
and Fur
so this
is
with their
SRC
okay
back to
here
we're in
implementations
there's a
guides
implementations
you know
all these
folder structures
can be
evolved
this is all
just kind
of like
inbox
as we

learn some
emerging
practices
for what's
happening
here
but like
we will
do more
and different
and better
so everyone
is welcome
to learn
and contribute
and raise
issues in
github
and do
more
so here
it is
is the
docs
here is
where I
copied in
the SRC
so they
already had
this open
source repo
with documentation
I brought
it in
then over
in
cursor

as always
just like
resynced
the cursor
repo
just to
double check
this was
when I
was doing
this
and
prompted
something
like
here
make an
obsidian
linked
readme
just to
help
totally
understand
rx
and fur
so here's
their
summary
and that
summary
exists
in
our
repo
too
but now
here we

are
with the
obsidian
linked
style
with all
these
different
loopy
belief
propagation
we've
been
there
and
then
I
kept
several
some
of
these
code
examples
especially
it's
like
let us
test
them
make
sure
that
they
work
and
so
on

what's
what's
a big
topic
in
rx
infer
what's
a big
motivator
well
doing
reactive
message
passing
so here's
the
reactive
programming
data
sources
reactive
layer
and then
the
inference
layer
core
concepts
stream
processing
patterns
error
handling
performance
considerations
these are
all plain

text
these are
mermaid
mind
maps
so
update
frequency
we can
just add
timing
and now
timing
is right
there
and it's
then when
we push
to github
it'll be
updated
because this
is just
a markdown
and then
when it's
published
to obsidian

sync
it'll just
be there
as part
of the
diagram
message
passing
in rx

infer
so just
each
time
I was
like
huh
what would
that wiki
article
be
or
what
would
that
knowledge
node
be
so let's
just look
through them
we have
active
inference
examples
and then
follow along
in red
with where
okay so
here we
are
active
inference
examples
were linked
out to
of course

the big
heavy
active
inference
one over
here
that's
probably
real
active
inference
yep
and then
a bunch
of other
ones
around
here
so
these
are
especially
if these
can be
validated
and verified
that they
really run
as we
expect them
to with
the app
models
that would
be
awesome
but these
are just

oh
what are
active
inference
examples
in rx
infer
now
they have
that
in the
examples
in their
documentation
so if
you
now we're
on their
website
examples
examples.rxinfer.ml
and then
here's like
the active
inference
mountain car
so
a good
cursor
a good
LLM
if we
said
make a
new
and we'll
just
let's just

do this
right now
just to
test
rxinfer
mountain
car
readme
dot
md
okay
new chat
here
write a
comprehensively
professional
readme
specifically
copying
and invoking
and building
upon
rxinfer
treatment
of the
classic
mountain
car
setting
use
obsidian
links
densely
ensure
that all
analytical
and rxinfer
syntax

threads
are
fully
unpacked
because
in the
docs
folder
in the
examples
it has
that full
mountain
car
it
has
the
mountain
car
but now
it's going
to write
an obsidian
style
readme
about
the
mountain
car
okay
advanced
systems
engineering
so this
was just
for systems
engineering
guidebook

like
the
LLMs
are trained
on systems
engineering
documents
just like
they're
trained
on a
variety
of other
technical
documents
so here's
different
stages
of model
design
and
architecture
and
lifecycle
design
from a
systems
engineering
perspective
and then
this is
the
advanced
systems
engineering
guidebook
I guess
okay

let's
look at
the
mountain
car
here's
the
rx
infer
mountain
car
readme
okay
that part
could
some stuff
again
it renders
really well
in github
but not
in local
obsidian
so
so you
know
but then
that's
where we
get the
meta
docs
that's
like
we
like
this
kind

of
linking
and
that
kind
if
it's
called
this
or
if
it
has
this
template
we
like
that
if
it's
like
that
we
like
this
okay
so we
could
check
does
this
this
looks
pretty
similar
does
that
actually

bring
in
oh
probabilistic
model
okay
and then
it's
like
we
already
have
that
page
but
if
we
didn't
we
could
say
write
that
page
go
back
inference
constraints
free
energy
components
active
inference
framework
optimize
the
expected
free

energy

let's

link

active

inference

there

double

bracket

active

inference

so now

it's a red

link

because we

already

have

that

page

policy

selection

oh

what is

policy

selection

it's

like

middle

click

how do

we think

about

policy

selection

in

active

inference

well

we

use
expected
free
energy
what are
the
expected
free
energy
components
well
one
decompositions
epistemic
and pragmatic
value
what
variables
used
to
apply
that
what
notation
or what
letters
it
associated
with
sometimes
with
the
e
and
so
okay
and
then

here's
the
running
like
this
just
generated
this
obsidian
style
plus
plus
plus
from
the
veridical
mountain
car
example
that the
developers
provided
yes
add
many
more
sections
as
relevant
including
an
intro
section
that
sets
up
a
wonderful

story
as to
why
we're
solving
it
this
way
okay
back
to the
rx
infer
docs
so
compiler
pipeline
pending
experts
and
others
questions
and
comments
but
this
is
very
fascinating
to
look
at
how
it
digs
into
the
documentation

and
the
code
itself
and
pulls
out
some
topics
that
even
in
the
year
plus
of
learning
about
this
package
we
haven't
necessarily
gone
too
much
into
detail
but
this
could
help
a lot
with
automated
code
production
and

improvement
and
also
onboarding
and
better
understanding
of
performance
all these
different
features
so
that's
the
compiler
pipeline
code
generation
all
right
accept
it
back
to
the
mountain
car
imagine
standing
at
the
bottom
of
a
steep
valley
with

an
underpowered
car
your
goal
reach
the
summit
the
catch
your
engine
alone
isn't
strong
enough
to
climb
directly
this
is
the
classic
mountain
car
problem
but
today
we're
going
to
solve
it
in
a
fundamentally
different
way

not
through
traditional
reinforcement
or
optimal
control
but
through
the
lens
of
active
inference
why
active
inference
and
it's
like
and
then
the
show
is
on
the
road
okay
execution
engine
of
rx
infer
here's
what it
looks like
over

in
github
that
top
section
looks like
that
in the
obsidian
program
locally
but
it looks
like
this
on
github
which
is
epic
execution
engine
message
flow
architecture
optimization
flow
message
passing
message
scheduling
cool
factor
graphs
key
topic
so
this

is a
display
kind
that
doesn't
work
in
the
github
but
let's
look
at
it
over
here
let's
see
if
it
looks
fine
okay
this
one
has
an
error
let's
see
if
it
can
fix
it
factor
graphs
we

are
getting
some
diagram
and
markdown
weirdness
and

failures
and
failures
and
failures
to
render
properly
on
github
please
expand
improve
obsidian
!

dial link
and ensure
total
comprehensive
representation
of
rx
and
for
technical
contents
within
properly
appearing

diagrams
and
etc.
okay
free
energy
how is
free
energy
playing a
role in
rx infer
how is it
calculated
that's what
this article
is
and then it's
like well
this is like
very technical
that's a good
thing
it serves as
a concise
linkable
place
and also
it can be
entered into
with learning
paths
write me a
learning path
to get
to
free
energy

in
policy
inference
as
planning
in
rx infer
for an
active
inference
agent
and
I'm
coming
from
this
perspective
how does
free
energy
play a
role
in
message
passing
here's
perception
belief
updating
with
variational
free
energy
and
then
action
inference
with

expected
free
energy
and
where
they
come
together
that's
epic
message
passing
and
active
inference
some
of
this
could
be
low
key
fabricated
but
having
the
rx
infer
docs
themselves
there
gives
a
scaffold
and
if it
was
ambiguous

enough
to
need
so much
interpolation
that's
a sign
to
improve
the
underlying
documents
and
to
improve
these
paradox
as
well
more
details
this
one I
just
said
write
a
follow
up
give
more
details
on
how
free
energy
plays
a

role
in
message
passing
here
okay
it made
the
improvements
to the
factor
graphs
article
let's
return
to
factor
graphs
this
part
still
looks
like
it's
an
error
let's
see
if
it
saved
it
push
it
to
github
okay
so

unable
to
render
rich
display
reload
it
new
version
just
happened
now
still
doesn't
work
but
people
kind
of
oh
wow
that's
so
cool
that's
factor
graph
style
um
getting
started
how do
you get
started
in
rx
infer
well

here's
conceptual
pre-reqs
here's
the
installation
this
looks
good
basic
concepts
your
first
model
coin
toss
that's
what we
did
too
visualizing
using
plots
more
advanced
features
working
with
constraints
custom
distributions
this is
getting
into
some
pretty
advanced
examples

message
passing
the
concept
model
macro
this is
the
at
model
macro
paradigm
this is
because
this is
kind
of
the
core
interesting
syntax

that
opens
up
an
incredible
world
of
semantics
the
at
model
macro
automatically
converts
model
specifications

into
factor
graphs
it
uses
this
at
model
style
syntactically
to
develop
these
enormously
performant
reactive
message
passing
setups
which
can be
used
for
batch
learning
or
for
streaming
models
can have
constraints
applied
in any
which
way
and
enables
reactivity

to occur
with this
kind of
subscription
based
approach
ast
that
we could
explore
more
about
that
add
another
comprehensive
rx infer
documentation
page
fully linked
about
ast
abstract
syntax
trace

it's
kind of
an
arcane
topic
but
it
matters
model
specification
here's how

you specify
a generative
model in
rx infer
here's
fanfiction
interactive
documentation
about how
you do
that
rx
infer
stack
these are
some
visualizations
that I
don't even
know
if they've
existed
before
new ways
and not
necessarily
even
accurate
ones
but
sometimes
on the
path
towards
utility
and
learning
of

the
model
specification
through the
app
model
macro
here's
the
ast
transformation
that's
the
abstract
syntax
tree
we're
about
to
look
at
that
page
that
is
being
written
right
now
yes
add
vast
comprehensive
sections
to
ast
page
with

analytical
details
diagrams
connections
okay
so
the
app
model
is
specified
syntax
wise
it's
transformed
using
this
abstract
syntax
tree
from
that
ast
transformation
the
factor
graph
is
compiled
rules
for
message
passing
are
compiled
optimization
passes
are

performed
we could
look in
more
detail
what
are
this
then
with
the
message
rules
just
in
time
prescribed
and
the
optimization
passes
of some
kind
we could
explore
situations
where
they
are
and
aren't
coming
into
play
then
there's
this
execution

engine
that
actually
runs
on
those
rails
as
constrained
and
specified
through
message
passing
node
local
message
passing
free
energy
minimization
and
there's
some
memory
management
too
okay
what
is
ast
parses
the
code
extracts
the
variables
identifies

the
dependencies
generates
a
graph
optimization
eliminates
dead
codes
merges
optimizations
and
optimizes
memory
now
we'll
accept
the
ast
article
let's
jump
over
to
this
new
ast
processing
article
abstract
syntax
trees
play a
crucial role
in
model
specification
within

rx
infern
the
at
model
macro
transforms
julio code
into
an
ast
which is
then
processed
to
generate
factor
graphs
for
probabilistic
inference
ast
type
system
is
based
on
type
theory
principles
here's
type
theoretic
treatment
of
ast
processing
here's

how it
connects
to
julia
type
abstract
distribution
continuous
or
discrete
two
examples
of
continuous
distributions
normal
Gaussian
and the
gamma
message
passing
on
ast
memory
management
in
ast
and
here's
how
variational
inference
is done
in
rx
infer
here's
streaming

inference

in

rx

infer

okay

um

oh

let's

look at

one quick

code

viz

okay

meanwhile

um

that

completes

our

primary

agenda

let's

reload

this

window

um

at

this

time

if

anyone

has

any

requests

or

ideas

write

it

and

i
will
stick
around
for a
few
more
minutes
otherwise
i hope
this
has
been
interesting
useful
possibly
relevant
um
um
let's
here
um
code
viz
is
doing
its
own
indexing
all
right
let's
do
okay
okay
i'll
just
make

one
um
okay
thank you
reconfigurability
okay
moonere
has
written
one
suggestion
i'm
just
going to
show
how
that
how
i'm
doing
that
there's
i
i know
more
and better
and different
and relevant
for you
um
okay
this is
like
wait a
minute
like
where
is

everything
okay
this is
going to
be
active
inference
okay
and
then
um
we'll
just call
that
um
path
for a
second
when
you're
wrote
i have
another
topic
experience
as a
method
by
carl
popper
and
the
least
action
principle
free
energy
minimization

theory
of
mind
i'm
curious
to
see
the
output
as
a
learning
path
all
right
we'll
call
this
experience
as a
method
or
the
learning
path
active
inference
experience
do we
want to
make
this
like
jimmy
hendrix
style
experience
method

path
develop
this
learning
path
with the
stated
topics
comprehensively
including
some
jimmy
hendrix
like
stylings
quote
are
you
experienced
okay
check out
codeviz
okay
still
the
codeviz
sometimes
takes a
while
i know
that they're
aware of
that
let's
just see
if it
gets
there

by the
end
of the
stream
okay
add
comprehensively
obsidian
linking
to
extant
files
and
karl
popper
philosophy
everywhere
possible
active
inference
pragmatism
learning
path
dot md
this is
kind of
fun
here's
just
doing
tab
autocomplete
style
and you
can get
do we
want to
complete

that one
i don't
know if
it's
the time
but we'll
okay
so let's
do
the
paparian
okay
that's
the
experience
method
one
here
in the
pragmatism
path
write
vastly
and
comprehensively
related to
american
pragmatism
and
operationalism
okay
let's look
at the
are you
experienced
learning
path
are you

experienced

a journey

through

active

inference

and

scientific

method

knowledge

speaks

but

wisdom

listens

jimi

hendrix

i may

be wrong

and you

may be

right

and by

an effort

we may

get

nearer

to the

truth

carl

popper

okay

experience

as method

walking

through the

purple

haze of

knowledge

carl

popper's
epistemological
revolution
practical
experiments
problem
tentative
solution
error
elimination
new
problem
yes
add more
information
on the
history
of
american
pragmatism
then
that we're
going to
queue up
this next
one
returning
to
the
experience
method
add a
lengthy
section
contrasting
the
paperian
hendrix

epistemology

with the

hegelian

synthetic

dialectic

method

and

kant

okay

we'll return

to that

one after

let's do

little

pragmatism

metaphysical

club

absolutely

cambridge

beginnings

holmes

right

james

peirce

emerson

classical

period

1878

1920

peirce

theory of

signs

james

the golden

age

1920

to

1940

mid-century
transitions
logical
empiricism
contemporary
renaissance
interesting
foundations
of thought
operationalism
and scientific
practice
bridgman
indeed
it really
was that
way
whoa
neo
operationalism
cognitive
operations
I'm sure
cognitive
security
active
through the
pragmatic
lens
social
dimension
so cool
so cool
okay
okay let's
go back to
the experience
as method

see if
see if that
hegelian
kantian
element
has a
jimi
hendrix
song
that
reflects
it
okay
philosophical
crossroads
bold as
love
critical as
truth
are you
experienced
have you
ever been
experienced
well I
have
jimi
hendrix
experience
experience
is not
what
happens
to you
it's
what
you
do

with
what
happens
to
you
carl
popper
whoa
hegelian
dialectic
the purple
haze of
spirit
kantian
foundation
transcendental
structures
paparian
revolution
electric
experience
synthesis
and contrast
the wind
cries
method
wind
cries
merry
integration
with active
inference
all along
the watch
tower
kantian
categories
predictive

models
hegelian
development
hierarchical
learning
preparing
and testing
error
correction
the story
of life
is quicker
than the
wink
of an
eye
jimi
hendrix
the growth
of knowledge
proceeds
from old
problems
to new
problems
carl
popper
remember
like
hendrix's
mastery
of the
guitar
and
popper's
pursuit
of truth
understanding

these
principles
comes
through
practice
experimentation
and bold
exploration
let your
mind
be free
to experience
predict
and learn
while maintaining
critical
thinking
okay
here
we'll just
do one
final
one
okay
ensure

that this
active
inference
so
I'm
going to
type
this
out
generate
it

probably
read
it
if it
looks
funny
and
interesting
then I'll
read any
final comments
in the live
chat
and then that
will be
the end
ensure that
this active
inference
learning
path
weaves
weaves
delightfully
around the
who's on
first
famous dialogue
the
topics of
active
inference
and
free energy
principle
should be
comprehensively
and technically

conveyed
amidst
a
rarefied
soup
of
baseball
lore
timeless
comedy
wonderful
wonderful
science
and the
occasional
appearance
by the
ghost of
John Nash
what
wonderful
dialogue
intergenerational
family
dynamics
and deep
philosophical
questions
of identity
agency
and the
nature
of reality
as
interpreted
in a
question
question

20s
golden
quote
golden
era
of
American
culture
in
technical
terms
that
are
accessible
to a
wide
audience
ensure
detailed
baseball
lore
and
active
inference
all
along
the
every
way
new
chat
make
this
happen
structure
the
learning
path

itself
as
who's
on
first
not
just
in
semantic
referent
!
vastly
expand
vastly
expanding
this
learning
path

ensure
technical
math
sections
and topics
and diagrams
and obsidian
!
linking
long

long
long
conversation
conveying
so much
lore
and

information

comma

it

really

does

boggle

the

minds

yeah

cursor

indexing

works

really

well

code

viz

not

quite

there

for

large

databases

yet

okay

thank

you

Jeff

thank

you

Andrew

Muneer

yeah

we'll see

where this

gets us

if it's

like

funny

and
interesting
possibly
read it
then
that'll
be it
welcome
to a
unique
okay
active
inference
learning
path
who's
on
first
introduction
the
perfect
game
of
mind
welcome
to a
unique
exploration
of active
inference
and the
free
energy
principle
structured
as the
immortal
who's

on
first
comedy
routine
just as
Abbott and
Costello's
routine
plays with
language and
expectations
will use
their format
to unpack
the fascinating
world of
how brains
make
reality
there we
are
for
those who
like a
little
topological
side
with their
baseball
comedy
all right
okay
I don't
think I'm
going to
read this
whole
thing

but it's
clearly
funny
let's
just see
what
happens
act
one
tell me
about
active
inference
okay
brains
predictions
technical
interlude
first base
you're
saying
there's
someone
called
prediction
on
first
who's
on
first
I'm
asking
you
who's
making
the
predictions
deep

dive
second
base
strategic
play
historical
baseball
analogy
just as
Bill
James
revolutionized
baseball
with
sabermetrics
Carl
Fristin
transformed
neuroscience
with a
free
energy
principle
that
should
be
an
obsidian
link
okay
whoa
sensory
input
prediction
who
free
energy
what

action
I don't
know
it's
like
the
OODA
loop
but
for
comedy
but
for
the
1920s
but
for
baseball
but
for
future
audiences
third
base
action
selection
the
outfield
extended
concepts
okay
little
bit
of
dialogue
little
bit
of

chatter
little
bit
of
peanut
gallery
practice
drills
who's
making
predictions
predictive
processing
exercise
what's
the
free
energy
mathematical
workshops
I don't
know
is on
policy
selection
here we
go
historical
context
the 1920s
golden
era
baseball
statistics
meets
brain
statistics
that

is
super
cool
add
many
more
connections
about
baseball
statistics
and
brain
statistics
give
phd
dissertation
level
and
length
about
how
statistical
parametric
mapping
SPM
for
sages
even
the future
of advanced
baseball
analytics
SPM
could be
posterior
probability
max
for where

the ball
is pitched
or hit
baseball
statistics
meets
brain
statistics
Babe Ruth's
batting
average
as a
probability
distribution
Miller
Huggins
strategy
as policy
selection
and the
1927
Yankees
as a
predictive
processing
machine
cultural
integration
advanced
topics
the perfect
game
statistical
bullpen
notes
for
learners
okay

whoa
see it
just
changed
while
we're
in
it
okay
now
here's
the
statistical
SPM
meets
sabermetrics
!
Hey
Abbott
these
numbers
are making
my head
spin
that's
because
you're
looking at
them like
box
scores
you need
to think
about
them like
brain
maps
Gaussian

random
field
theory
in
baseball
and
brains
it
literally
is
that
way
advanced
analytics
the
convergence
spatial
normalization
and player
normalization
literally
it is
that
way
temporal

basis
functions
pitch
sequencing
effects
fatigue
modeling
!
statistical
!
methods

it's like
I feel like
people are
ready for
SPM
applied to
baseball
but
baseball
applied to
SPM
is
evergreen
okay I'll read
this last comment
and then any other
live chats
and then it's over
remember
just as Abbott and
Costello's routine
seems confusing at
first but has a
clear internal
logic
active inference
might initially
appear complex
but follows
coherent principles
keep practicing
and soon you'll be
running the bases
of cognitive
science
like a pro
ending of
number

0

10.2

push the

github

then

in

Obsidian

push the

new

and the

updated

done

close

Obsidian

okay

ending of

10.2

close it

thank you

Claude

and so many

ways we could

continue from

there

but that's

where I'll

call it

for tonight

so I'm

glad that

the internet

held up

mostly well

enough

to make

this happen

looking forward

to where

people
take it
all from
here
so
till next
time
peace
to make

Thank you.